

Khumbulani Gumede

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EDUCATION

University of KwaZulu-Natal PMB, Scottville

B.S. in Computer Science and IT

Graduation, May 2025

- **Concentrations:** Computer Systems and AI.
- **Related Coursework:** Data Structures & Algorithms, Database Systems, Computer Systems, Machine Learning, Artificial Intelligence, Natural Language Processing, Object-Oriented Programming, Theory of Computation, Software Design, Graphics & Modelling

CERTIFICATIONS

TryHackMe – SOC Level 1 Learning Path *May 2026*

TryHackMe – DevSecOps Learning Path *Mar 2026*

TryHackMe – Web Application Red Teaming Learning Path *Feb 2026*

TryHackMe – Web Application Pentesting Learning Path *Dec 2025*

PROJECTS

FifaScope - Football Prediction Dashboard *May 2026*

Personal Project: Individual

- Built a full-stack football match prediction dashboard combining statistical machine learning with Generative AI (Google Gemini) for intelligent match analysis and insights.
- Designed a multi-database persistence layer separating live fixtures, active predictions, and archived match history, ensuring clean data integrity across the application lifecycle.
- Implemented a React frontend with a FastAPI backend, applying defensive programming patterns (optional chaining, strict team validation) to eliminate runtime errors and ensure production-grade stability.
- Architected an API-efficient workflow that runs bulk statistical predictions on sync and reserves calls for on-demand single-match enrichment, adhering to rate limits while maximising insight quality.

TradeScope - Trading Analytics Platform

Aug 2025

Personal Project: Individual

- Built a full-featured desktop trading analytics platform, featuring an interactive performance dashboard, calendar view, and MetaTrader 5 (MT5) data integration.
- Developed a dynamic trade replay system with animated candlestick visualisation (Plotly.js) and entry/exit markers, enabling traders to review historical setups and refine strategies.
- Implemented a rich-text journaling and notes workspace (TipTap editor) with local time tracking and file/screenshot attachments, providing a single interface for trade review and performance insights.
- Managed local SQLite databases for all trade, note, and OHLC candle data, prioritising privacy and performance.

MyFitness GPT

Aug 2024 – Oct 2024

Software Design Course: Team Member

- Developed a web-based personalised fitness advisor that generates user-specific health profiles and delivers tailored guidance through GPT-powered chatbot interactions.
- Applied MVC architecture to ensure clean separation of concerns across the Model (user fitness profiles), View (web UI), and Controller (data processing and chatbot integration).
- Overcame challenges in personalisation, PDF processing, and chatbot accuracy, achieving an 86% project grade.

Emerland Cipher Game

Aug 2024 – Oct 2024

Graphics & Modelling Course: Team Member

- Co-developed a maze-runner puzzle game set in an ancient Scottish castle, featuring riddle solving, isometric gameplay, and AI-driven enemies with escalating difficulty.
- Contributed to level design, interactive storytelling, and game mechanics, achieving a 78% project grade.

ACTIVITIES AND LEADERSHIP

University of KwaZulu-Natal

Demonstrator for Comp201 (Data Structures & Algorithms)

Aug 2024 – Dec 2024

- Supported a class of 60 students during practical sessions in Java-based Data Structures and Algorithms.
- Provided one-on-one assistance with assignments, test preparation, and exam revision to strengthen student understanding of core course material.

SKILLS

Programming Languages: Java, Python, C++, C#, JavaScript, TypeScript

Web & Application Frameworks: Spring Boot, React, Express (Node.js)

Frontend Technologies: TailwindCSS, Electron, Plotly.js, Recharts, Vite, TipTap

Databases: MySQL, SQLite

Machine Learning & AI: XGBoost, Scikit-learn, Google Gemini API, OpenAI/GPT API, Jupyter Notebooks, Google Colab

Cybersecurity: Web Application Pentesting, Web Application Red Teaming, SOC Level 1 Analysis, DevSecOps Practices

Tools & IDEs: Git, Visual Studio Code, Eclipse, PyCharm, Unity, VS Code

Methodologies: MVC Architecture, REST API Design, Agile Development, Database Design, Object-Oriented Programming